Chapter 2 - The Real Number System

Problem Set # 2

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1. If $a, b \in \mathbb{R}$ and $b \neq 0$, show that:
   (a) $|a| = \sqrt{a^2}$
   (b) $|a/b| = |a|/|b|$
2. If $a, b \in \mathbb{R}$, show that $|a + b| = |a| + |b|$ if and only if $ab \geq 0$.
3. If $x, y, z \in \mathbb{R}$ and $x \leq z$, show that $x \leq y \leq z$ if and only if $|x - y| + |y - z| = |x - z|$. Interpret this geometrically.
4. Show that $|x - a| < \varepsilon$ if and only if $a - \varepsilon < x < a + \varepsilon$.
5. If $a < x < b$ and $a < y < b$, show that $|x - y| < b - a$. Interpret this geometrically.
6. Find all $x \in \mathbb{R}$ that satisfy the following inequalities:
   (a) $|4x - 5| \leq 13$
   (b) $|x^2 - 1| \leq 3$
7. Find all $x \in \mathbb{R}$ that satisfy the equation $|x + 1| + |x - 2| = 7$.
8. Find all values of x that satisfy the following equations:
   (a) $x + 1 = |2x - 1|$
   (b) $2x - 1 = |x - 5|$
9. Find all values of x that satisfy the following inequalities. Sketch graphs.
   (a) $|x - 2| \leq x + 1$
   (b) $3|x| \leq 2 - x$
10. Find all $x \in \mathbb{R}$ that satisfy the following inequalities:
    (a) $|x - 1| > |x + 1|$
    (b) $|x| + |x + 1| < 2$

11. Sketch the graph of the equation $y = |x| - |x - 1|$.

12. Find all $x \in \mathbb{R}$ that satisfy the inequality $4 < |x + 2| + |x - 1| < 5$.

13. Find all $x \in \mathbb{R}$ that satisfy both $|2x - 3| < 5$ and $|x + 1| > 2$ simultaneously.

14. Determine and sketch the set of pairs (x, y) in $\mathbb{R} \times \mathbb{R}$ that satisfy:

    (a) $|x| = |y|$

    (b) $|x| + |y| = 1$

    (c) $|xy| = 2$

    (d) $|x| - |y| = 2$

15. Determine and sketch the set of pairs (x, y) in $\mathbb{R} \times \mathbb{R}$ that satisfy:

    (a) $|x| \leq |y|$

    (b) $|x| + |y| \leq 1$

    (c) $|xy| \leq 2$

    (d) $|x| - |y| \geq 2$

16. Let $\varepsilon > 0$ and $\delta > 0$, and $a \in \mathbb{R}$. Show that $V_\varepsilon(a) \cap V_\delta(a)$ and $V_\varepsilon(a) \cup V_\delta(a)$ are γ-neighborhoods of a for appropriate values of γ.

17. Show that if $a, b \in \mathbb{R}$, and $a \neq b$, then there exist ε-neighborhoods U of a and V of b such that $U \cap V = \emptyset$.

18. Show that if $a, b \in \mathbb{R}$ then

    (a) $\max\{a, b\} = \frac{1}{2}(a + b + |a - b|)$ and $\min\{a, b\} = \frac{1}{2}(a + b - |a - b|)$

    (b) $\min\{a, b, c\} = \min\{\min\{a, b\}, c\}$

19. Show that if $a, b, c \in \mathbb{R}$, then the "middle number" is

$$\text{mid}\{a, b, c\} = \min\{\max\{a, b\}, \max\{b, c\}, \max\{c, a\}\}$$

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"The essence of mathematics is not to make simple things complicated, but to make complicated things simple." - Stan Gudder -